Day-33 Jan-21

IPv6 Part-3

IPv6 representation.

*) Leading 0s must be removed.
2001:0db8:0000:0001:0f2a:4fff:fea3:00b7
↳ 2001:db8:0:1:f2a:4fff:fea3:b7

*) :: Must be used to shorten the longest string of all-0 quartets (if there is only one all-0 quartets, don't use '::')
Eg:
2001:0000:0000:0f2a:0000:0000:00b7.
↳ 2001::f2a:0:0:b7

*) If there are 2 equal-length choices for the ::, use :: to shorten the one on the left.
2001:0db8:0000:0000:0f2a:0000:0000:00b1
↳ 2001:0db8::0f2a:0:0:b7

* Hex-characters (a-f) must be written using lowercase, Not uppercase A B C D E F.

IPv6 Header.

Offsets	Octet	0				1				2				3			
Octet	Bit	0 1 2 3	4 5 6 7	8 9 10 11	12 13 14 15	16 17 18 19	20 21 22 23	24 25 26 27	28 29 30 31								
0	0	Version	Traffic Class			Flow Label											
4	32	Payload Length				Next Header				Hop Limit							
8-12	64-96	Source Address															
16-20	128-160																
24	192	Destination Address															
28	224																
32-36	256-288																

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IPv6 has a fixed size of 40 bytes i.e 320 bits
The processing of IPv6 header is simpler for router. So, performance is improved.

Version
- Length: 4 bits.
- Indicates the version of IP that is used.
- Fixed value of 6 (0b0110) to indicate IPv6.

Traffic Class.
- Length: 8 bits
- Used for QOS (Quality of Service), to indicate high-priority traffic
- For eg - IP phone traffic, live video calls etc. to will have a Traffic Class value which gives them priority over other traffic.

Flow-Label
- Length- 20 bits.
- Used to identify specific traffic 'flows' (communication between specific source & destination)

Payload-Length.
- 16 bits in length
- Indicates the length of Payload (the encapsulated Layer 4 segment in bytes)
- The length of IPv6 header itself isn't included, because its always 40 bytes i.e fixed.

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Next-Header.

- Length - 8 bits
- Indicates the type of the next-header (head of the encapsulated segment), for example TCP or UDP.
- Same fxn as the IPv4 header's Protocol field

Hop Limit

- Length: 8 bits
- The value of this field is decremented by 1 by each router that forwards it. If it reaches, the packet is discarded.
- Same fxn as the IPv4 header's 'TTL' field

Source/Destination

- Length: 128 bits each.
- These fields contain the IPv6 address of the packet's source and the packet's intended destination.

NOB: IPv6 header bits qsth anuney chance less hunta?

Solicited-Node Multicast address

- An IPv6 solicited-node multicast address is calculated from a unicast address.

FF02:0000:0000:0000:0000:0001:ff + Last 6 digit of unicast address

Eg:

2001:0db8:0000:0001:0f2a:4ff:fea3:00b1 - unicast address

↓

ff02::1:ffa3:b1

2001:0db8:0000:0001:0489:4eda:073a:12b8 - unicast address

↓

ff02::1:ff3a:12b8

(Tyo show ipv6 int g0/0 garda Joined Group bhnye hunxa, tesko last ma yesto ta solicited node address)

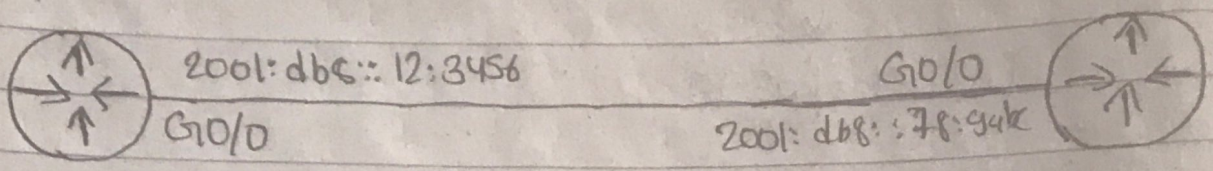
Neighbor Discovery Protocol

- Neighbor Discovery Protocol (NDP) is a protocol used with IPv6.
- It has various functions, and one of these functions is to replace ARP, which is no longer used in IPv6.
- The ARP-like fx" of NDP uses ICMPv6 and solicited-node multicast addresses to learn the MAC address of other hosts. (* ARP in IPv4 uses broadcast)
- Two message types are ~~sent~~ used:

1) Neighbor Solicitation (NS) = ICMPv6 Type 135.
2) Neighbor Advertisement (NA) = ICMPv6 Type 136.

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Neighbor Solicitation (NS)



Suppose, we tried to Ping R2 G0/0 from R1. For that it needs to know R2 mac. How will it do it? For this it uses NS mssg.

For this it sends Hello mssg to R2. with src, dst ip & mac as follows

- Source: R1 G0/0 IP.
- Destination: IP: R2 solicited-node multicast address.
  - (it knows soliated add. from dst IP cmmnd)
- Source MAC: R1 G0/0 MAC.
- Dst MAC: Multicast MAC based on R2's solicited-node address.

  Ether, Dst MAC: IPv6 mcast - ff:78:99:bc.

Difference between NDP NS mssg & ARP is ~~NDP uses~~ that ARP is broadcast & NS is multicast (more efficient)

Neighbor Advertisement.

NDP equivalent of ARP reply.

- R2 received a mssg from R1 req its MAC add.
- In response R2 just sends R1 its MAC.

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IPv6 Neighbor Table

- All devices using IPv6 keep neighbor table instead of arp table as in IPv4

R1 # show ipv6 neighbor.

IPv6 address	Age	Link-layer Addr	State	Interface.

Age: Indicates how long ago R1 received traffic from these addresses in mins.

Link layer Add: Shows R2's MAC.add.

Neighbor Discovery Protocol.

- Another functions of NDP allows hosts to automatically discover routers on the local network.

- Two messages are used for this ~~prop~~ proces.

1). Router Solicitation (RS) = ICMPv6 Type 133
   - ↳ Sent to Multicast address: FF02::2 (all routers)
   - ↳ Asks all routers on the local link to identify themselves.
   - ↳ Sent when an interface is enabled / host is connected to the network.

2)- Router Advertisement (RA) = ICMPv6 Type 134.
   - ↳ Sent To multicast address FF02::1 (all nodes).
   - → The routers announces its presence, as well as other information about the link.
   - → These messages are sent in response to RS messages
   - ↳ They are also sent periodically, even if the router hasn't received an RS

SLAAC

- Stands for Stateless Address Auto-Configuration.
- Hosts uses the RS/RA messages to learn the IPv6 prefix of the local link (ie 2001:db8::/64).
- Using the ipv6 address prefix/prefix-length eui-64 command, you need to manually enter the prefix.
- Using the ipv6 address autoconfig command, you don't need to enter the prefix. The device uses NDP, to learn the prefix on the local link.
- The device will use EUI-64 to generate the interface ID, a it will be randomly generated (depending on device(maker).

  Example:

  R1 is connected with R2, R1 is configured with ipv6 but R2 ain't.

```
R2(config)# int g0/0
R2(config-if)# ipv6 address autoconfig
R2(config-if)# do show ipv6 int brief
```

Duplicate Address Detection (DAD).

- One final point about NDP.
- Duplicate Address Detection (DAD) allows hosts to

check if other devices on the local link are using the same IPv6 address.

- Any time an IPv6-enabled interface initializes (no shutdown command) or an IPv6 address is configured on an interface (by any method: manual, SLAAC etc), it performs DAD

- DAD uses two messages we learned earlier: NS and NA
- The host will send an NS to its own IPv6 address. If it doesn't get a reply, it knows the address is unique.
- If it gets reply, it means another host on the network is already using the address.

IPv6 Static Routing.

- IPv6 routing works the same as IPv4 routing.
- However, the two processes are separate on the router, and the two routing tables are separate as well.

- IPv4 routing is enabled by default.
- IPv6 routing is disabled by default, and must be enabled with ipv6 unicast-routing.

- If IPv6 routing is disabled, the router will be able to send & receive IPv6 traffic, but won't route IPv6 traffic (= won't forward it between networks)
  $ show ipv6 route ↵ - Command.
- A connected network route is automatically added for each connected network.
- A local host route is automatically added for each address configured on the router.

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- Routes for link local address aren't added to routing table.

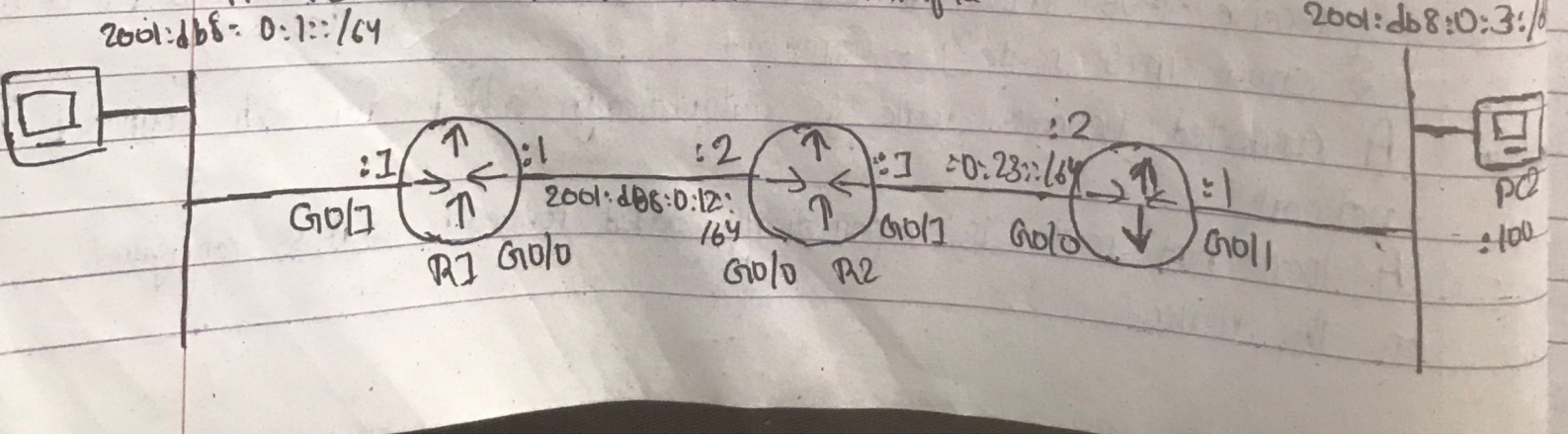
Command for IPv6 static routing:
ipv6 route destination/prefix-length {next-hop | exit-interface [next-hop]} [ad]

*) Directly attached static route:
Only the exit interface is specified.
ipv6 route destination/prefix-length exit-interface
R1(config)# ipv6 route 2001:db8:0:3::/64 g0/0
(ethernet को case मा कम्पल्सरी हुन्छ)

*) Recursive static route: Only the next-hop is specified
ipv6 route destination/prefix-length next-hop
R1(config)# ipv6 route 2001:db8:0:3::/64 2001:db8:0:12::2

*) Fully specified static route:
Both exit interface and next-hop are specified.
ipv6 route destination/prefix-length exit-interface next-hop
R1(config)# ipv6 route 2001:db8:0:3::/64 g0/0 2001:db8:0:12::2

NOB:
In IPv6, we can't use directly attached static routes if the interface is an Ethernet interface.



#) Network Route:

R1(config)# ipv6 route 2001:db8:0:3::/64 2001:db8:0:12::2

#) Host Route:

R2(config)# ipv6 route 2001:db8:0:1::100/128 2001:db8:0:12::1

R2(config)# ipV6 route 2001:db8:0:3::100/128 2001:db6:0:23:

#) Default Route.

R3(config)# ipV6 route ::/0 2001:db8:0:23::1.

::2

3::/64

PC2

:100